Chapter 10. Regression Diagnostics

Hat matrix and leverage, residuals, deletion formulas, influence measures

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10.1 What can go wrong

Fitting $\mathbf{y} = X\beta + \epsilon$, $\epsilon \sim N(\mathbf{0}, \sigma^2 I)$, by $\hat\beta = (X'X)^{-1}X'\mathbf{y}$ is justified only if the model is adequate. **Regression diagnostics** look for:

(1) violations of the error assumptions (normality, independence, equal variance), studied through residuals;

(2) a wrong functional form, remedied by transforming y or the x_j (Chapters 5 and 11);

(3) near-singularity of $X'X$ (multicollinearity, Chapter 15);

(4) **high-leverage points**: observations whose $\mathbf{x}_i$ is far from the bulk of the design;

(5) **outliers**: observations whose y_i is far from its fitted value;

(6) **influential observations**: observations whose deletion changes the fit substantially.

Points of types (4)–(6) need not coincide: a high-leverage point can lie on the line, and an outlier at the centre of the design may hardly move the fit.

10.2 The hat matrix and leverage

The fitted values are $\hat{\mathbf{y}} = H\mathbf{y}$ with $H = X(X'X)^{-1}X' = (h_{ij})$, where $h_{ij} = \mathbf{x}_i'(X'X)^{-1}\mathbf{x}_j$ and $\mathbf{x}_i' = (1, x_{i1}, \dots, x_{ip})$ is the i-th row of X.

Theorem 10.1 (Properties of the hat matrix)

(a) H is symmetric and idempotent with $\operatorname{rank} H = \operatorname{tr} H = \sum_i h_{ii} = p + 1$.

(b) $h_{ii} = \sum_j h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2$; hence $0 \le h_{ii} \le 1$ and $|h_{ij}| \le \tfrac{1}{2}$ for $i \neq j$.

(c) $HX = X$. If the model has an intercept, $H\mathbf{1} = \mathbf{1}$, i.e. every row of H sums to 1, and $h_{ii} \ge 1/n$.

(d) With an intercept, $h_{ii} = \frac{1}{n} + (\mathbf{x}_i^* - \bar{\mathbf{x}})'(\tilde X'\tilde X)^{-1}(\mathbf{x}_i^* - \bar{\mathbf{x}})$, where $\mathbf{x}_i^* = (x_{i1}, \dots, x_{ip})'$ and $\tilde X$ is the centred $n \times p$ matrix of regressors. For $p = 1$, $h_{ii} = \frac{1}{n} + \frac{(x_i - \bar x)^2}{S_{xx}}$.

Proof. (a) $H' = H$, $H^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = H$, and $\operatorname{tr} H = \operatorname{tr}\{(X'X)^{-1}X'X\} = p + 1$. (b) The (i,i) entry of $H = HH'$ is $\sum_j h_{ij}^2$. Since it is at least h_{ii}^2, $h_{ii} \ge h_{ii}^2$, so $0 \le h_{ii} \le 1$. For $j \neq i$, $h_{ij}^2 \le h_{ii} - h_{ii}^2 \le \frac{1}{4}$. (c) $HX = X$ is direct; the intercept column gives $H\mathbf{1} = \mathbf{1}$. For $h_{ii} \ge 1/n$ see (d). (d) Reparametrize with centred regressors: the column space of X is spanned by $\mathbf{1}$ and the columns of $\tilde X$, which are orthogonal to $\mathbf{1}$. Hence $H = \frac{1}{n}\mathbf{1}\mathbf{1}' + \tilde X(\tilde X'\tilde X)^{-1}\tilde X'$, and the diagonal gives the formula, the second term being nonnegative. For $p = 1$, $\tilde X'\tilde X = S_{xx}$. ∎

Since $\hat y_i = \sum_j h_{ij} y_j$, the weight of y_i in its own fitted value is $h_{ii} = \partial \hat y_i / \partial y_i$; h_{ii} is the **leverage** of observation i. By (d), it grows with the (Mahalanobis) distance of $\mathbf{x}_i^*$ from the centroid $\bar{\mathbf{x}}$: with $M(i) = (\mathbf{x}_i^* - \bar{\mathbf{x}})'\{\tilde X'\tilde X/(n-1)\}^{-1}(\mathbf{x}_i^* - \bar{\mathbf{x}})$,

$$M(i) = (n-1)\Big(h_{ii} - \frac{1}{n}\Big).$$

The average leverage is $\bar h = (p+1)/n$, and a common rule flags $h_{ii} > 2\bar h$ (or $3\bar h$) as high leverage.

10.3 Residuals

Theorem 10.2

$\mathbf{e} = (I-H)\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2(I-H))$. So $\operatorname{Var} e_i = \sigma^2(1-h_{ii})$, $\operatorname{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$, and $\operatorname{Corr}(e_i, e_j) = -h_{ij}/\sqrt{(1-h_{ii})(1-h_{jj})}$.

High-leverage points have small residual variance: the fit is pulled towards them. To compare residuals on a common scale:

Definition (Scaled residuals)

With $s^2 = MSE = \mathbf{e}'\mathbf{e}/(n-p-1)$ and $s^2_{(i)}$ the MSE computed without observation i:

- the **standardized** (internally studentized) residual is $r_i = \dfrac{e_i}{s\sqrt{1-h_{ii}}}$;
- the **studentized** (externally studentized) residual is $r_i^* = \dfrac{e_i}{s_{(i)}\sqrt{1-h_{ii}}}$.

r_i is not t-distributed, because e_i and s are dependent (a large $|e_i|$ inflates s). Deleting observation i removes the dependence. Let $X_{(i)}$, $\mathbf{y}_{(i)}$ denote X, $\mathbf{y}$ without row i, $\hat\beta_{(i)}$ the corresponding estimator, and $\tilde y_i = \mathbf{x}_i'\hat\beta_{(i)}$ the prediction of y_i from the other observations.

Theorem 10.3 (Deletion formulas)

(a) $\hat\beta_{(i)} = \hat\beta - \dfrac{(X'X)^{-1}\mathbf{x}_i e_i}{1-h_{ii}}$.

(b) $y_i - \tilde y_i = \dfrac{e_i}{1-h_{ii}}$ (the **deleted** or PRESS residual).

(c) $(n-p-2)s^2_{(i)} = (n-p-1)s^2 - \dfrac{e_i^2}{1-h_{ii}}$.

Proof. (a) $X'_{(i)}X_{(i)} = X'X - \mathbf{x}_i\mathbf{x}_i'$ and $X'_{(i)}\mathbf{y}_{(i)} = X'\mathbf{y} - \mathbf{x}_i y_i$. With $A = X'X$, the Sherman–Morrison formula gives

$$(A - \mathbf{x}_i\mathbf{x}_i')^{-1} = A^{-1} + \frac{A^{-1}\mathbf{x}_i\mathbf{x}_i'A^{-1}}{1-h_{ii}}.$$

Multiplying by $X'\mathbf{y} - \mathbf{x}_i y_i$ and using $A^{-1}X'\mathbf{y} = \hat\beta$, $\mathbf{x}_i'\hat\beta = \hat y_i$, $\mathbf{x}_i'A^{-1}\mathbf{x}_i = h_{ii}$:

$$\hat\beta_{(i)} = \hat\beta - A^{-1}\mathbf{x}_i y_i + \frac{A^{-1}\mathbf{x}_i(\hat y_i - h_{ii}y_i)}{1-h_{ii}} = \hat\beta + \frac{A^{-1}\mathbf{x}_i\{\hat y_i - h_{ii}y_i - (1-h_{ii})y_i\}}{1-h_{ii}} = \hat\beta - \frac{A^{-1}\mathbf{x}_i e_i}{1-h_{ii}}.$$

(b) $\tilde y_i = \mathbf{x}_i'\hat\beta_{(i)} = \hat y_i - h_{ii}e_i/(1-h_{ii})$, so $y_i - \tilde y_i = e_i + h_{ii}e_i/(1-h_{ii}) = e_i/(1-h_{ii})$. (c) For $j \neq i$, the deleted residual is $y_j - \mathbf{x}_j'\hat\beta_{(i)} = e_j + h_{ji}e_i/(1-h_{ii})$. Summing squares over $j \neq i$ and using $\sum_j e_j h_{ji} = 0$ (since $H\mathbf{e} = \mathbf{0}$) and $\sum_j h_{ji}^2 = h_{ii}$:

$$SSE_{(i)} = \sum_{j\neq i} e_j^2 + \frac{2e_i}{1-h_{ii}}\sum_{j\neq i} h_{ji}e_j + \frac{e_i^2}{(1-h_{ii})^2}\sum_{j\neq i} h_{ji}^2.$$

Here $\sum_{j\neq i} e_j^2 = SSE - e_i^2$, $\sum_{j\neq i} h_{ji}e_j = -h_{ii}e_i$, and $\sum_{j\neq i} h_{ji}^2 = h_{ii} - h_{ii}^2$. So

$$SSE_{(i)} = SSE - e_i^2 - \frac{2h_{ii}e_i^2}{1-h_{ii}} + \frac{h_{ii}e_i^2}{1-h_{ii}} = SSE - \frac{e_i^2}{1-h_{ii}}.$$

■

Theorem 10.4 (Distribution of the studentized residual)

(a) $r_i^* = \dfrac{y_i - \tilde{y}_i}{s_{(i)}\sqrt{1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i}} \sim t(n-p-2)$.

(b) $r_i^* = r_i\sqrt{\dfrac{n-p-2}{n-p-1-r_i^2}}$, a monotone function of r_i.

Proof. (a) y_i is independent of $\hat{\beta}_{(i)}$ and $s^2_{(i)}$, which are computed from the other $n-1$ observations; so $y_i - \tilde{y}_i \sim N(0, \sigma^2\{1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i\})$, independent of $(n-p-2)s^2_{(i)}/\sigma^2 \sim \chi^2(n-p-2)$. By Sherman–Morrison, $1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i = 1 + h_{ii} + h_{ii}^2/(1-h_{ii}) = 1/(1-h_{ii})$, and with Theorem 10.3(b) the ratio equals $e_i/\{s_{(i)}\sqrt{1-h_{ii}}\}$. (b) Substitute Theorem 10.3(c), $s^2_{(i)} = s^2(n-p-1-r_i^2)/(n-p-2)$. ∎

So an observation is flagged as an outlier when $|r_i^*| > t_{\alpha/2}(n-p-2)$. Since n tests are performed, a Bonferroni level α/n is safer.

10.4 Influence measures

An observation is influential if deleting it changes the fitted model a lot. The measures below combine the residual and the leverage.

Theorem 10.5 (Influence measures)

(a) **DFFITS**: $\mathrm{DFFITS}_{(i)} = \dfrac{\hat{y}_i - \tilde{y}_{i(i)}}{s_{(i)}\sqrt{h_{ii}}} = \left(\dfrac{h_{ii}}{1-h_{ii}}\right)^{1/2} r_i^*$, where $\tilde{y}_{i(i)} = \mathbf{x}_i'\hat{\beta}_{(i)}$.

(b) **Cook's distance**: $D_{(i)} = \dfrac{(\hat{\beta} - \hat{\beta}_{(i)})'X'X(\hat{\beta} - \hat{\beta}_{(i)})}{(p+1)s^2} = \dfrac{r_i^2}{p+1}\cdot\dfrac{h_{ii}}{1-h_{ii}}$.

(c) **COVRATIO**: $\mathrm{COVRATIO}_{(i)} = \dfrac{\det\{s^2_{(i)}(X_{(i)}'X_{(i)})^{-1}\}}{\det\{s^2(X'X)^{-1}\}} = \dfrac{1}{(1-h_{ii})}\left(\dfrac{n-p-2+r_i^{*2}}{n-p-1}\right)^{-(p+1)}$.

(d) **Andrews–Pregibon**: with $Z = (X, \mathbf{y})$, $\mathrm{AP}_{(i)} = \dfrac{\det(Z_{(i)}'Z_{(i)})}{\det(Z'Z)} = 1 - h_{ii} - \dfrac{e_i^2}{SSE}$.

Proof. (a) By Theorem 10.3(a), $\hat{y}_i - \tilde{y}_{i(i)} = \mathbf{x}_i'(\hat{\beta} - \hat{\beta}_{(i)}) = h_{ii}e_i/(1-h_{ii})$; divide by $s_{(i)}\sqrt{h_{ii}}$. (b) $(\hat{\beta} - \hat{\beta}_{(i)})'X'X(\hat{\beta} - \hat{\beta}_{(i)}) = \frac{e_i^2}{(1-h_{ii})^2}\mathbf{x}_i'(X'X)^{-1}\mathbf{x}_i = \frac{h_{ii}e_i^2}{(1-h_{ii})^2}$; divide by $(p+1)s^2$. (c) $\det(X_{(i)}'X_{(i)}) = \det(X'X)\,(1-h_{ii})$ by the matrix determinant lemma, so the ratio is $(s^2_{(i)}/s^2)^{p+1}/(1-h_{ii})$. By Theorem 10.3(c), $s^2_{(i)}/s^2 = (n-p-1-r_i^2)/(n-p-2)$, and by Theorem 10.4(b), $n-p-2+r_i^{*2} = (n-p-2)(n-p-1)/(n-p-1-r_i^2)$. Hence $s^2_{(i)}/s^2 = (n-p-1)/(n-p-2+r_i^{*2})$, which gives the formula. (d) By the determinant lemma, $\det(Z_{(i)}'Z_{(i)})/\det(Z'Z) = 1 - \mathbf{z}_i'(Z'Z)^{-1}\mathbf{z}_i$, where $\mathbf{z}_i' = (\mathbf{x}_i', y_i)$. The last quantity is the leverage of observation i in the regression on Z, which by the partitioned-inverse formula equals $h_{ii} + e_i^2/SSE$. ∎

Rules of thumb (Belsley, Kuh and Welsch; Cook):

- $|\mathrm{DFFITS}_{(i)}| > 2\sqrt{(p+1)/n}$;
- $D_{(i)} > F_{0.50}(p+1, n-p-1)$ (roughly 1): deleting i moves $\hat{\beta}$ outside the 50% confidence ellipsoid of Theorem 8.2;

- $|\text{COVRATIO}_{(i)} - 1| > 3(p+1)/n$;
- small $\text{AP}_{(i)}$.

Every measure grows with both $|r_i|$ and h_{ii}: influence is "outlyingness $\times$ leverage".

Example 10.1 (Two kinds of unusual point)

In a classical data set of $n = 21$ children (age at first word x, aptitude score y), the fit is $\hat{y} = 109.87 - 1.13x$. Observation 18 has $x = 42$, far from the others, with $h_{18,18} = 0.65 > 3\bar{h} = 0.29$: a high-leverage point. Its residual is small ($r^* = -0.85$), yet $D_{(18)} = 0.68$ and $\text{DFFITS}_{(18)} = -1.16$ are the largest: it is influential. Observation 19 has the largest residual, $r^*_{19} = 3.61 > t_{0.025}(18) = 2.10$: an outlier, but with ordinary leverage ($h = 0.053$), so its Cook's distance is only 0.22. The two points call for different actions: check the measurement of the outlier, and assess whether the conclusions depend on the single high-leverage point.

What to do. An unusual point is first checked for recording errors. If it is genuine, it may signal a wrong model (a missing variable or nonlinearity); robust regression or reporting fits with and without the point are better than silently deleting it. Influence diagnostics based on deleting one point at a time can miss groups of points that mask each other.